COLLEGE OF COMPUTING SCIENCES AND INFORMATION TECHNOLOGY

TEERTHANKER MAHAVEER UNIVERSITY, MORADABAD

INTERNSHIP SYNOPSIS

ON

SECURE FILE VAULT – A HYBRID FILE ENCRYPTION & TAMPER DETECTION SYSTEM

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ABSTRACT

In today's digital environment, sensitive files stored on personal systems are increasingly exposed to risks such as unauthorized access, data theft, corruption, and tampering. To address these challenges, this project introduces **Secure File Vault**, a desktop-based cybersecurity application designed to protect important files through encryption, integrity verification, and secure access control.

The system allows users to create an authenticated account and manage files through a simple graphical interface. Using the **Advanced Encryption Standard (AES)** with **PBKDF2 key derivation**, the application converts normal files into secure encrypted files that cannot be opened without the correct password. To maintain file integrity, the system uses **SHA-256 hashing**, which acts as a digital fingerprint. Any change in the file—intentional or accidental—results in a different hash value, enabling the application to detect tampering instantly.

A key highlight of this project is its **hybrid database system**, which ensures smooth functioning both online and offline. When internet connectivity is available, the application connects to **MongoDB Atlas**, offering cloud-based accessibility and secure remote storage. In offline conditions, it automatically switches to **Local MongoDB**, allowing uninterrupted use without requiring network access. This design reflects modern industry practices where hybrid cloud architectures are commonly used for reliability and flexibility.

The user interface, developed using **Python Tkinter**, provides a clean and responsive dashboard for encryption, decryption, file verification, and log viewing. All major activities—such as login, logout, file encryption, decryption, and integrity checks—are recorded as logs, ensuring accountability and easy monitoring. Additional features like automatic session timeout enhance overall security by preventing misuse during inactivity.

Overall, **Secure File Vault** combines practical cybersecurity techniques with a user-friendly interface to create a reliable tool for personal and professional file protection. The project demonstrates how encryption, hashing, authentication, and hybrid data management can be effectively implemented using open-source technologies. It also highlights the importance of accessible security tools that help users protect their digital information with ease.

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INTRODUCTION TO SECURE FILE VAULT

In today's digital era, a large amount of personal, academic, and professional information is stored on computers, laptops, and external storage devices. With this rapid growth of digital data, the risks associated with cyber threats have also increased significantly. Users often face issues like unauthorized access, data theft, malware, corruption, and accidental file tampering. Even a small unwanted change in an important file can cause serious problems in fields such as education, finance, business, healthcare, and government services. Although people are aware of these risks, they usually do not have access to simple and reliable tools that can protect their important files without requiring advanced technical knowledge. This creates the need for a user-friendly application that ensures confidentiality, integrity, and secure access to sensitive data.

To address this requirement, the **Secure File Vault** application has been developed as a practical cybersecurity solution. It provides users with a safe and convenient way to encrypt files, verify their integrity, and protect them from unauthorized modifications. The system uses the **Advanced Encryption Standard (AES)** along with **PBKDF2-based key derivation** to convert normal files into encrypted formats that cannot be opened without the correct password. Even if someone gains access to the encrypted file, its content remains unreadable and secure, giving users confidence when storing sensitive documents.

To ensure that files remain original, the application also includes an integrity verification mechanism using **SHA-256 hashing**. This hashing method generates a unique digital fingerprint for every file. If a file is altered—even by one character—the hash value changes instantly. By comparing the stored hash with the current one, the system can easily detect tampering or corruption. This feature is highly valuable for maintaining the authenticity of important documents.

One of the key strengths of this project is its **hybrid database system**, which allows the application to operate smoothly in both online and offline environments. When the internet is available, the system connects to **MongoDB Atlas**, a cloud-based database for securely storing user accounts, file metadata, salts, and logs. When offline, it automatically switches to **Local MongoDB**, ensuring uninterrupted access and functionality. This hybrid approach increases reliability and makes the system suitable for real-world use.

The user interface, built with **Python Tkinter**, provides a clean and interactive dashboard where users can register, log in, add files, encrypt or decrypt them, verify integrity, and view activity logs. Features such as password strength checking, secure password recovery, and automatic session timeout further enhance security and make the application easy to use.

Overall, the **Secure File Vault** project demonstrates practical cybersecurity principles in a simple and accessible way, offering an effective solution for everyday file protection.

FEATURES OF SECURE FILE VAULT

1. Advanced File Encryption (AES):
The system secures files using AES encryption combined with PBKDF2-derived keys, turning normal files into unreadable ciphertext. This ensures confidential data stays protected even if copied or accessed by unauthorized users.

2. Secure File Decryption:
Encrypted files can be restored to their original form using the correct password. Only the authorized user can decrypt the file, keeping private information safe and controlled.

3. Tamper Detection with SHA-256:
Each file is assigned a unique SHA-256 hash to serve as a fingerprint. If the file is modified or corrupted, the hash changes, allowing the system to instantly detect tampering.

4. Hybrid Database Mode:
The app automatically switches between MongoDB Atlas (online) and Local MongoDB (offline). This ensures uninterrupted access whether the user has internet or not, making the system reliable in any situation.

5. Secure User Authentication:
Passwords and security answers are protected using PBKDF2 hashing, making them extremely difficult to crack. Only verified users can access the vault, ensuring strong access control.

6. Activity Logging:
All user actions—login, logout, encryption, decryption, integrity checks—are recorded in logs. These logs help track usage and maintain transparency inside the vault.

7. Simple and Secure GUI Dashboard:
Built with Python Tkinter, the interface is clean, user-friendly, and easy to navigate. Users can add, encrypt, decrypt, and verify files without needing technical knowledge.

8. Auto-Lock After Inactivity:
To prevent unauthorized access, the system automatically locks itself when left idle. The user must log in again to continue, adding an extra layer of security.

9. Multi-File Management:
Users can add and manage multiple files inside the vault. Each file displays its status, hash, and timestamp, making organization and tracking smooth and efficient.

10. Background Thread Processing:
Heavy operations like encryption and decryption run in background threads. This prevents the app from freezing and ensures a smooth, responsive user experience.

SYSTEM ARCHITECTURE

1. Client-Side Graphical Interface (Tkinter):

The system provides a clean, responsive desktop interface built using Python Tkinter. Users can log in, register, add files, view logs, and perform encryption or decryption easily. The GUI is designed to be simple enough for non-technical users while maintaining strong security.

2. Authentication and Access Control Layer:

This layer verifies user credentials using PBKDF2-hashed passwords and hashed security answers. Only authorized users can access the vault. It ensures secure login, password recovery, and session handling without exposing sensitive information.

3. Encryption & Decryption Engine (AES + PBKDF2):

All encryption operations are handled using AES with keys derived through PBKDF2. This engine converts files into secure encrypted formats and restores them when needed. Heavy operations run in background threads to keep the system smooth and responsive.

4. File Integrity Verification Module (SHA-256):

Each file is assigned a unique SHA-256 hash when added to the vault. During integrity checks, the system recalculates the hash and compares it with the stored one. Any mismatch instantly reveals tampering or corruption.

5. Hybrid Database Architecture:

The system uses two databases—MongoDB Atlas (cloud) and Local MongoDB (offline). It connects to Atlas when the internet is available and switches to the local database when offline. This ensures continuous functionality regardless of network conditions.

6. Data Storage Layer:

All user data, file metadata, salts, encryption status, and activity logs are securely stored in the selected database. File contents are not stored; only metadata and cryptographic information are saved, ensuring privacy and minimizing storage load.

7. Activity Logging System:

Every important action such as login, logout, encryption, decryption, file addition, and integrity checks is recorded in the logs collection. This traceability helps users track usage and identify any suspicious activity.

8. Auto-Lock Security Mechanism:

The system includes an inactivity tracker that monitors user actions. If no activity is detected for a predefined period, the vault automatically locks and returns to the login screen, preventing unauthorized access.

9. Background Task Manager:

Time-consuming operations like encryption, decryption, and hashing run on separate threads. This prevents the GUI from freezing and ensures that users can continue interacting with the application smoothly.

10. System Workflow Controller:

This component manages communication between the GUI, security modules, and database layers. It ensures data flows correctly and all components work together seamlessly, maintaining both reliability and security.

COMPANY PROFILE

Codec Technologies Pvt. Ltd. is a well-established IT training and development organization committed to providing high-quality technical education and practical industry-oriented learning. The company focuses on reducing the gap between theoretical knowledge learned in academics and the practical skills required in the professional IT world. With experienced trainers and a structured training environment, Codec Technologies has become a trusted name for students and professionals looking to build careers in modern technologies.

Codec Technologies specializes in various domains such as:
- Cybersecurity and Ethical Hacking
- Cloud Computing (AWS, Google Cloud, Azure)
- Python Programming and Machine Learning
- Full-Stack Web Development
- Data Science and Big Data Analytics
- Networking and Server Administration

These programs are designed to provide both conceptual understanding and real-time practical exposure. The company regularly updates its curriculum to match current industry standards and technological advancements.

An important part of Codec Technologies is its internship and industrial training programs, where students get the opportunity to work on live projects under professional guidance. Through these internships, learners gain practical experience with real tools, workflows, and project environments. This helps in developing technical confidence, problem-solving ability, and teamwork skills—qualities essential for success in the IT industry.

The organization offers a well-equipped learning environment with modern computer labs, high-speed internet, expert trainers, and access to required software tools. Codec Technologies also conducts workshops, seminars, and competitive coding sessions to encourage creative thinking and innovation among students.

In addition to training, the company collaborates with various IT firms and industry partners, helping provide placement support and career guidance to deserving candidates. These collaborations keep the institute aligned with the latest industry requirements and ensure students receive relevant and updated knowledge.

Overall, Codec Technologies Pvt. Ltd. provides a professional and supportive learning environment that helps students build strong technical foundations and prepares them for successful careers in the IT field. It continues to contribute significantly to skill development through its comprehensive training programs and quality-driven approach.

CONCLUSION

The **Secure File Vault** project successfully demonstrates how essential cybersecurity principles can be applied to protect sensitive digital information in a simple and user-friendly manner. In an era where data theft, unauthorized access, and file tampering have become common threats, this application offers a practical solution through strong encryption, integrity verification, and secure access control. By implementing AES encryption with PBKDF2 key derivation, the system ensures that files remain confidential and protected, while the SHA-256 hashing mechanism helps maintain the authenticity and originality of stored data.

One of the major strengths of this project is its **hybrid database structure**, which allows seamless operation in both online and offline environments. The integration of MongoDB Atlas for cloud storage and Local MongoDB for offline mode makes the application reliable, flexible, and suitable for users who may not always have consistent internet access. This approach reflects real-world industry scenarios where hybrid systems are used to ensure continuous functionality and data availability.

The user interface designed with Python Tkinter provides an interactive and accessible platform for performing tasks such as file encryption, decryption, integrity checks, and activity tracking. Users can easily manage multiple files, view detailed logs, and benefit from features like auto-lock, password strength detection, and secure password recovery. The background threading mechanism also ensures smooth performance by preventing the application from freezing during heavy operations.

Overall, the Secure File Vault system meets its objective of providing a secure, efficient, and practical method for protecting important files. It brings together different elements of cybersecurity—including encryption, hashing, authentication, and secure workflow management—into a single application that is both powerful and easy to use. Although built as an academic project, its real-world relevance, reliability, and technical design reflect modern security standards. The project not only enhances understanding of cybersecurity concepts but also demonstrates how these techniques can be effectively applied to solve everyday data protection challenges.